2001 ESCAPE DRIVESHAFT REPLACEMENT AND REAR DIFFERENTIAL LEAK INSPECTION/REPAIR

AFFECTED VEHICLES: **2001 MODEL YEAR ESCAPE 4X4**

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OVERVIEW

This program combines two procedures, a driveshaft replacement and a differential leak inspection/repair. These two procedures are closely related, and performing them together will save time.

DRIVESHAFT REPLACEMENT:

All affected vehicles will have a ***new*** driveshaft installed.

DIFFERENTIAL LEAK INSPECTION/REPAIR:

The mass damper is a weight bolted to the right side of the differential assembly. The damper may have been installed using the wrong tightening sequence. When installed incorrectly the damper does not contact all three mounting pads as the bolts are tightened. This creates a leak in the silicone seal between the rotary blade coupling housing and the differential housing. An inspection will be performed to identify leaking differential assemblies. If a leak is found, the coupling housing will be removed and resealed with silicone (F7TZ-19554-AA). The damper will be reinstalled using the correct tightening sequence.

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INSPECTION

1. Raise and support the vehicle.
2. Wipe clean the seam between the rotary blade coupling and the differential housing. Check for fluid seepage at the seam. See Figure 1.
   - If no signs of fluid are found at the seam, perform Driveshaft Replacement only.
   - If signs of fluid are found at the seam, perform Driveshaft Replacement and Differential Reseal and Mass Damper Re-Installation.

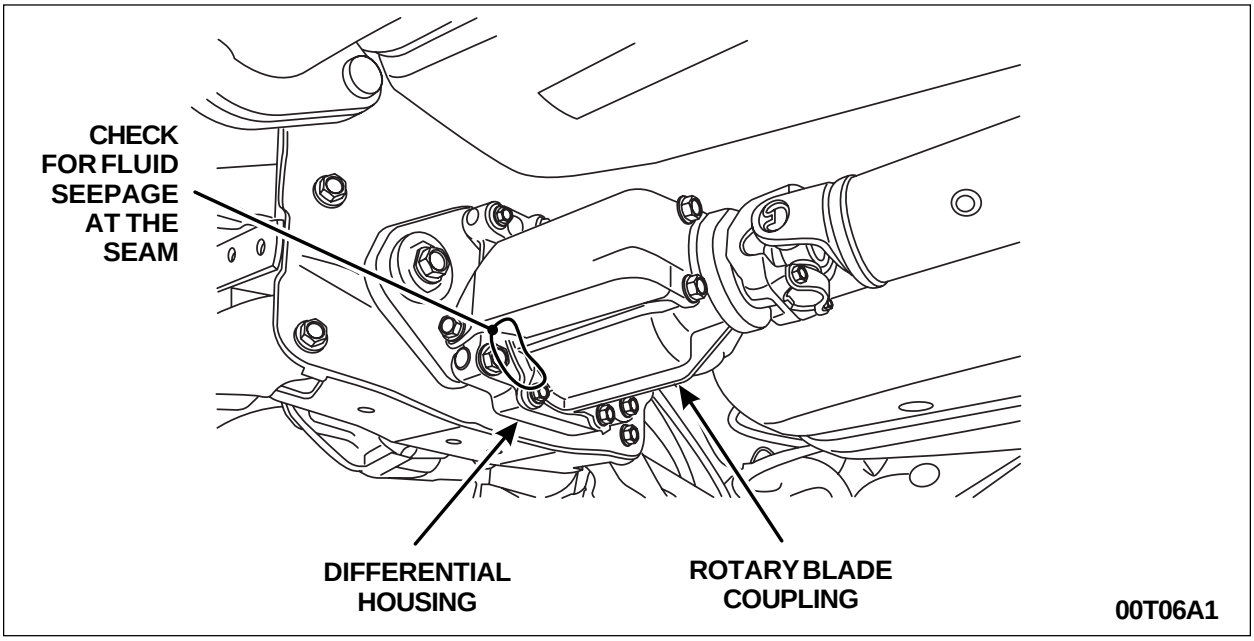


FIGURE 1

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DRIVESHAFT REPLACEMENT

CAUTION: Do not reuse any driveshaft retaining fasteners. Install *new* fasteners only or damage to vehicle may occur.

1. Position a safety stand under the driveshaft center bearing support. See Figure 2.

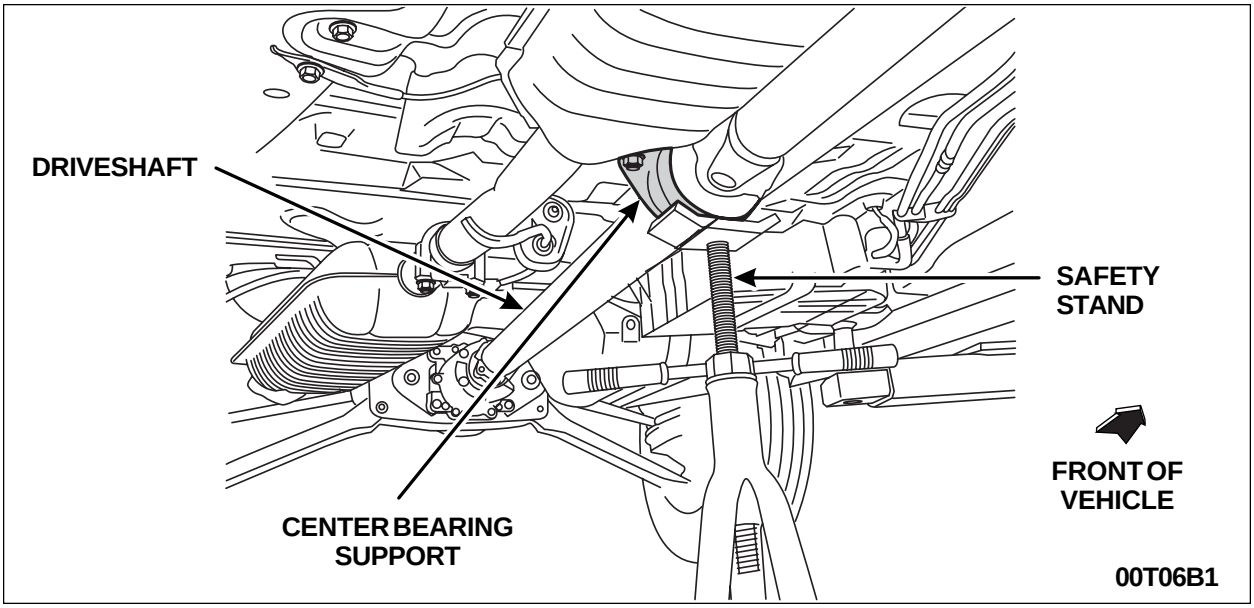


FIGURE 2

2. Remove the bolts and straps securing the rear of the driveshaft to the differential yoke. Discard the bolts. See Figure 3.

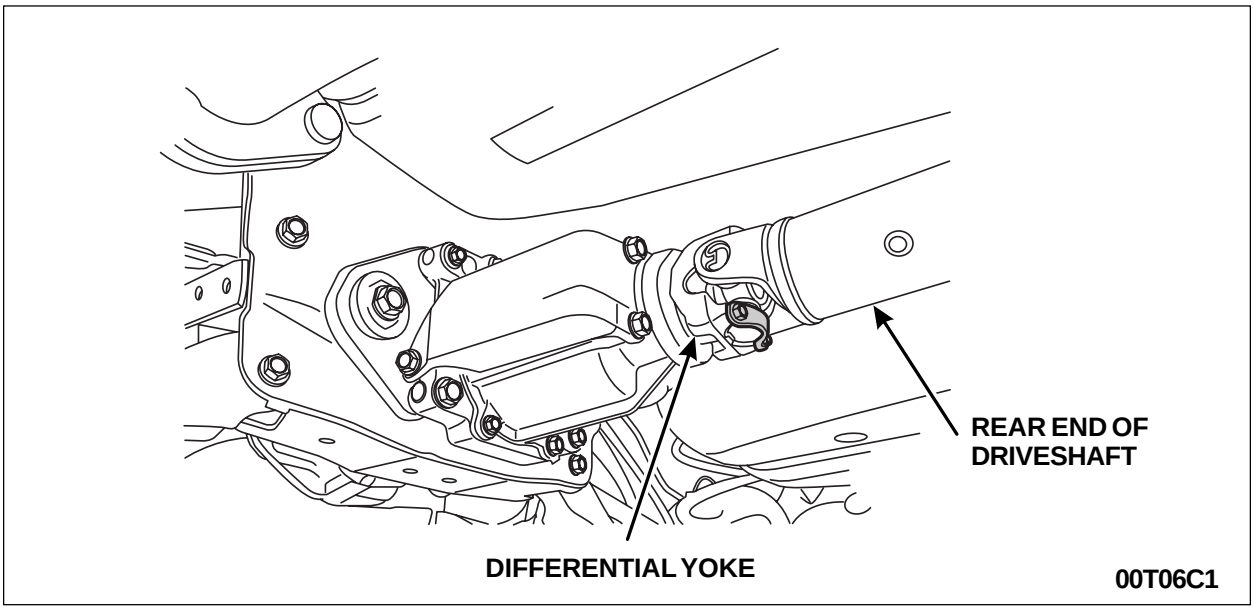


FIGURE 3

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3. Remove and discard the center bearing support nuts. See Figure 4.
4. Remove the center bearing support ground strap bolt from the floor pan. See Figure 4.

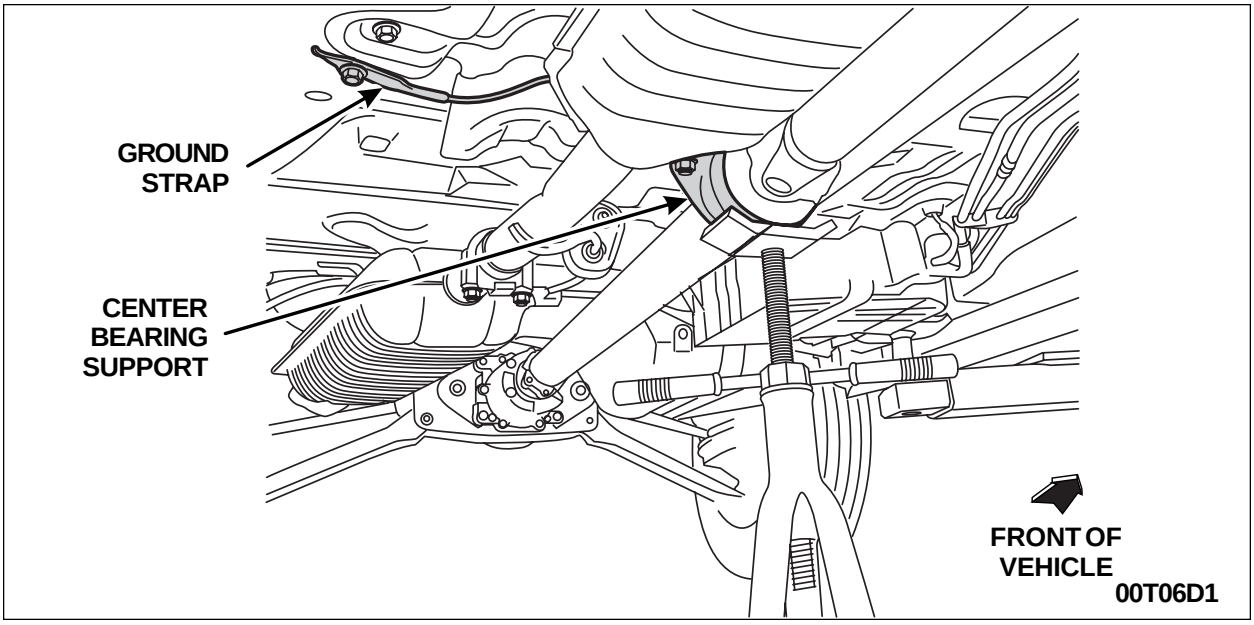


FIGURE 4

5. Remove and discard the bolts and washers securing the front of the driveshaft to the power take off. See Figure 5.

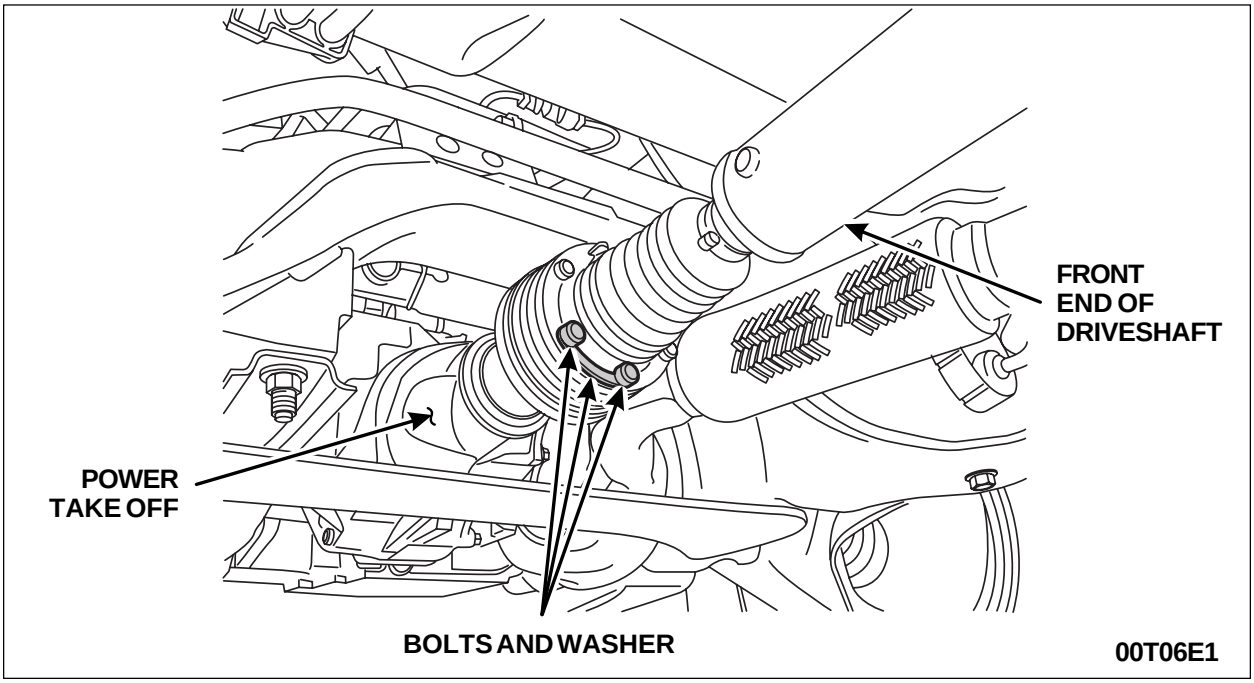


FIGURE 5

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6. With the aid of an assistant, remove the driveshaft from the safety stand. Discard the driveshaft.



IF REQUIRED, PERFORM DIFFERENTIAL RESEAL AND MASS DAMPER RE-INSTALLATION *NOW* BEFORE INSTALLING THE NEW DRIVESHAFT. GO TO PAGE 6.

7. With the aid of an assistant, position the center bearing support of the ***new*** driveshaft onto the safety stand. Position the rear of the driveshaft into the differential yoke, then position the front of the driveshaft into the power take off.
8. **CAUTION: The rear of the driveshaft must be bolted to the differential yoke before the center bearing support nuts are installed.**

   Install the straps with ***new*** bolts to secure the rear of the driveshaft to the differential yoke. Tighten the bolts to 23 Nm (17 lb-ft).
9. Install ***new*** nuts to the center bearing support. Tighten the nuts to 48 Nm (35 lb-ft).
10. Install the center bearing support ground strap to the floor pan. Tighten the bolt to 40 Nm (30 lb-ft).

NOTE: If front driveshaft bolts are missing or damaged, order Part Number YL8Z-4529-AA, quantity of one (1).

11. Install ***new*** bolts and washers to secure the front of the driveshaft to the power take off. Tighten the bolts to 37 Nm (27 lb-ft).
12. Remove the safety stand from under the center bearing support.
13. Lower the vehicle.

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DIFFERENTIAL RESEAL AND MASS DAMPER RE-INSTALLATION

1. Disconnect the electrical connector from the rotary blade coupling.
2. Remove the mass damper. See Figure 6.

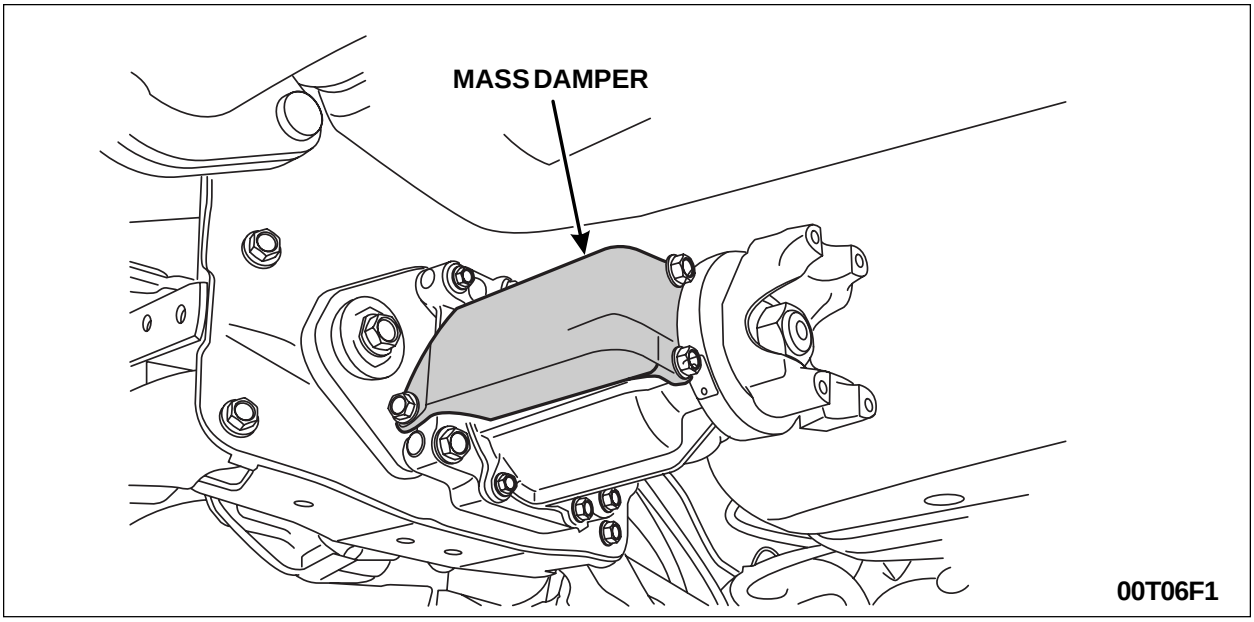


FIGURE 6

3. Position a drain pan under the differential assembly.
4. Remove the six bolts, then remove the rotary blade coupling. See Figure 7.

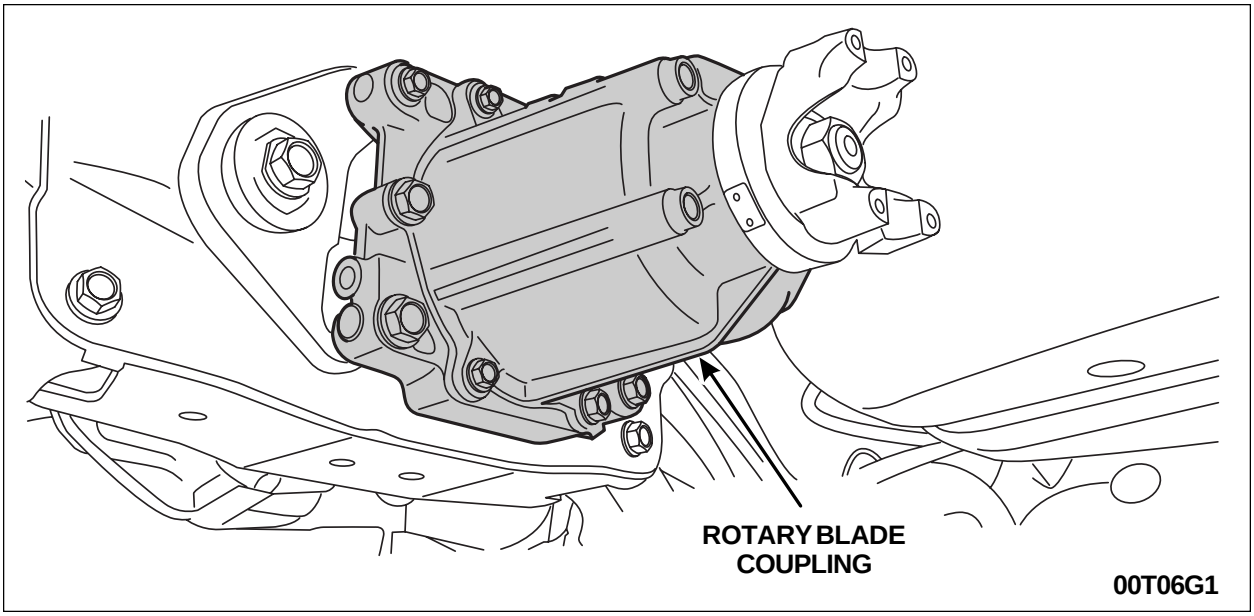


FIGURE 7

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5. **CAUTION: Make sure the mating surfaces of the rotary blade coupling and the differential housing are clean and free of oil before applying silicone sealant. Do not use abrasive cleaning pads. The inside of the differential housing must be covered when cleaning to prevent debris from entering the differential.**

   Clean the gasket mating surfaces of the rotary blade coupling and the differential housing with Ford Metal Surface Cleaner (F4AZ-19A536-RA).

6. NOTE: The rotary blade coupling must be installed within five (5) minutes of applying the silicone, or ***new*** silicone must be applied. If possible, allow one (1) hour before filling with lubricant to allow proper curing of the silicone.

   Apply a continuous bead of silicone sealant (F7TZ-19554-AA) or equivalent to the mating surface of the rotary blade coupling. Make sure the bead covers 360 degrees around each bolt hole. Make sure the rubber wiring grommet **does not** touch or damage the ***new*** silicone bead.

7. NOTE: The rubber wiring grommet must be guided into the slot on the differential housing during installation of the rotary blade coupling.

   Install the rotary blade coupling. Tighten bolts to 32 Nm (24 lb-ft).

8. **CAUTION: The proper torque sequence must be followed when installing the mass damper. Improper sequence will create a leak at the seam between the differential housing and the rotary blade coupling.**

   CAUTION: The initial torque for the damper rear bolt should NOT exceed 4 Nm (35 LB-INCHES).

   Install the mass damper and finger tighten the bolts. Torque the damper as follows. See Figure 8.
   - **Step 1: Torque rear damper bolt to 4 Nm (35 LB-INCHES) MAXIMUM**.
   - Step 2: Torque front bottom damper bolt to 48 Nm (35 lb-ft).
   - Step 3: Torque front top damper bolt to 48 Nm (35 lb-ft).
   - Step 4: Torque rear bottom damper bolt to 48 Nm (35 lb-ft).

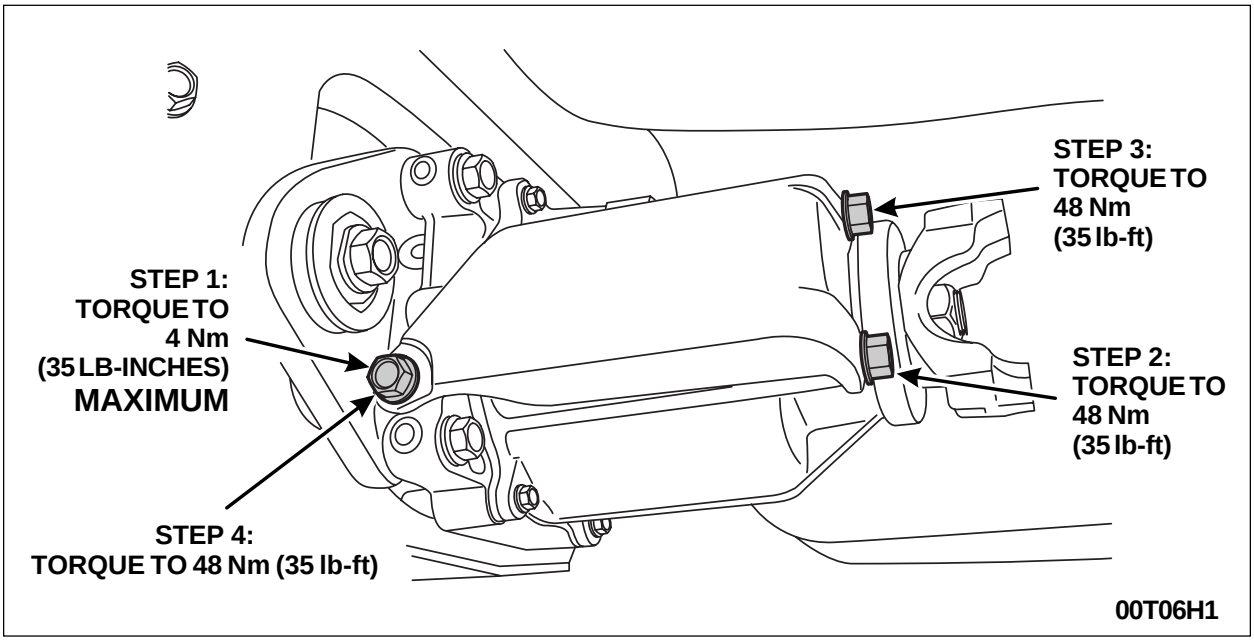


FIGURE 8

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9. Connect the electrical connector to the rotary blade coupling.
10. Remove the fill plug from the differential cover.
11. Fill differential with approximately 2 pints of XY-80W90-QL rear axle lubricant. Fluid level should be between 6 mm (1/4 inch) and 14 mm (9/16 inch) from bottom of filler plug opening.
12. Tighten the fill plug to 27 Nm (20 lb-ft).
13. Return to Step No.7 on Page 5, to complete driveshaft installation procedure.

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